

# R Basics

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## Load Packages

```
library(psych)
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2    3.5.1      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.1
## ✓ purrr      1.0.2
## — Conflicts — tidyverse_conflicts() —
## ✖ ggplot2::%>%() masks psych::%>%()
## ✖ ggplot2::alpha() masks psych::alpha()
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(nycflights13)
```

## Look at Data

```
glimpse(flights)
```

```
## Rows: 336,776
## Columns: 19
## $ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2...
## $ month     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
## $ day       <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
## $ dep_time  <int> 517, 533, 542, 544, 554, 554, 555, 557, 557, 558, 558, ...
## $ sched_dep_time <int> 515, 529, 540, 545, 600, 558, 600, 600, 600, 600, 600, ...
## $ dep_delay <dbl> 2, 4, 2, -1, -6, -4, -5, -3, -3, -2, -2, -2, -2, -2, -1...
## $ arr_time  <int> 830, 850, 923, 1004, 812, 740, 913, 709, 838, 753, 849,...
## $ sched_arr_time <int> 819, 830, 850, 1022, 837, 728, 854, 723, 846, 745, 851,...
## $ arr_delay <dbl> 11, 20, 33, -18, -25, 12, 19, -14, -8, 8, -2, -3, 7, -1...
## $ carrier   <chr> "UA", "UA", "AA", "B6", "DL", "UA", "B6", "EV", "B6", "...
## $ flight    <int> 1545, 1714, 1141, 725, 461, 1696, 507, 5708, 79, 301, 4...
## $ tailnum   <chr> "N14228", "N24211", "N619AA", "N804JB", "N668DN", "N394...
## $ origin    <chr> "EWR", "LGA", "JFK", "JFK", "LGA", "EWR", "EWR", "LGA", "...
## $ dest      <chr> "IAH", "IAH", "MIA", "BQN", "ATL", "ORD", "FLL", "IAD", "...
## $ air_time  <dbl> 227, 227, 160, 183, 116, 150, 158, 53, 140, 138, 149, 1...
## $ distance  <dbl> 1400, 1416, 1089, 1576, 762, 719, 1065, 229, 944, 733, ...
## $ hour      <dbl> 5, 5, 5, 5, 6, 5, 6, 6, 6, 6, 6, 6, 6, 6, 5, 6, 6, 6...
## $ minute    <dbl> 15, 29, 40, 45, 0, 58, 0, 0, 0, 0, 0, 0, 0, 0, 59, 0...
## $ time_hour <dtm> 2013-01-01 05:00:00, 2013-01-01 05:00:00, 2013-01-01 0...
```

Let's look at the first six rows of the data.

```
head(flights)
```

```
## # A tibble: 6 × 19
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>       <dbl>   <int>         <int>
## 1  2013     1     1     517           515         2     830           819
## 2  2013     1     1     533           529         4     850           830
## 3  2013     1     1     542           540         2     923           850
## 4  2013     1     1     544           545        -1    1004          1022
## 5  2013     1     1     554           600        -6     812           837
## 6  2013     1     1     554           558        -4     740           728
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>
```

Let's look at the first twenty rows of the data.

```
head(flights, 20)
```

```
## # A tibble: 20 × 19
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>       <dbl>   <int>         <int>
## 1  2013     1     1     517           515         2     830           819
## 2  2013     1     1     533           529         4     850           830
## 3  2013     1     1     542           540         2     923           850
## 4  2013     1     1     544           545        -1    1004          1022
## 5  2013     1     1     554           600        -6     812           837
## 6  2013     1     1     554           558        -4     740           728
## 7  2013     1     1     555           600        -5     913           854
## 8  2013     1     1     557           600        -3     709           723
## 9  2013     1     1     557           600        -3     838           846
## 10 2013     1     1     558           600        -2     753           745
## 11 2013     1     1     558           600        -2     849           851
## 12 2013     1     1     558           600        -2     853           856
## 13 2013     1     1     558           600        -2     924           917
## 14 2013     1     1     558           600        -2     923           937
## 15 2013     1     1     559           600        -1     941           910
## 16 2013     1     1     559           559         0     702           706
## 17 2013     1     1     559           600        -1     854           902
## 18 2013     1     1     600           600         0     851           858
## 19 2013     1     1     600           600         0     837           825
## 20 2013     1     1     601           600         1     844           850
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>
```

Let's look at the different variable types (aka the "kind" of variable: character, factor, numeric, integer, etc.).

```
class(flights)
```

```
## [1] "tbl_df"      "tbl"        "data.frame"
```

## Using filter()

select() all variables except carrier THEN filter() to only include observations whose destination is ORD.

```
flights %>%
  select(-carrier) %>%
  filter(dest == "ORD")
```

```
## # A tibble: 17,283 × 18
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
## 1  2013     1     1     554           558         -4     740           728
## 2  2013     1     1     558           600         -2     753           745
## 3  2013     1     1     608           600          8     807           735
## 4  2013     1     1     629           630         -1     824           810
## 5  2013     1     1     656           700         -4     854           850
## 6  2013     1     1     709           700          9     852           832
## 7  2013     1     1     715           713          2     911           850
## 8  2013     1     1     739           745         -6     918           930
## 9  2013     1     1     749           710         39     939           850
## 10 2013     1     1     828           830         -2    1027          1012
## # i 17,273 more rows
## # i 10 more variables: arr_delay <dbl>, flight <int>, tailnum <chr>,
## #   origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>, hour <dbl>,
## #   minute <dbl>, time_hour <dtm>
```

filter() all flights on January 1st.

```
flights %>% filter(month == 1, day == 1)
```

```
## # A tibble: 842 × 19
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
## 1  2013     1     1     517           515          2     830           819
## 2  2013     1     1     533           529          4     850           830
## 3  2013     1     1     542           540          2     923           850
## 4  2013     1     1     544           545         -1    1004          1022
## 5  2013     1     1     554           600         -6     812           837
## 6  2013     1     1     554           558         -4     740           728
## 7  2013     1     1     555           600         -5     913           854
## 8  2013     1     1     557           600         -3     709           723
## 9  2013     1     1     557           600         -3     838           846
## 10 2013     1     1     558           600         -2     753           745
## # i 832 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>
```

filter() all flights on January 1st and save into a new data frame.

```
jan1 <- flights %>% filter(month == 1, day == 1)
```

filter() all flights on December 25th, save into a new data frame, and print().

```
(dec25 <- flights %>% filter(month == 12, day == 25))
```

```
## # A tibble: 719 × 19
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
## 1  2013    12    25     456           500         -4     649           651
## 2  2013    12    25     524           515          9     805           814
## 3  2013    12    25     542           540          2     832           850
## 4  2013    12    25     546           550         -4    1022          1027
## 5  2013    12    25     556           600         -4     730           745
## 6  2013    12    25     557           600         -3     743           752
## 7  2013    12    25     557           600         -3     818           831
## 8  2013    12    25     559           600         -1     855           856
## 9  2013    12    25     559           600         -1     849           855
## 10 2013    12    25     600           600          0     850           846
## # i 709 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>
```

`filter()` all flights on December 25th, save into a new data frame, and `print()`.

```
(dec25 <- flights %>% filter(month == 12, day == 25))
```

```
## # A tibble: 719 × 19
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
## 1  2013    12    25     456           500         -4     649           651
## 2  2013    12    25     524           515          9     805           814
## 3  2013    12    25     542           540          2     832           850
## 4  2013    12    25     546           550         -4    1022          1027
## 5  2013    12    25     556           600         -4     730           745
## 6  2013    12    25     557           600         -3     743           752
## 7  2013    12    25     557           600         -3     818           831
## 8  2013    12    25     559           600         -1     855           856
## 9  2013    12    25     559           600         -1     849           855
## 10 2013    12    25     600           600          0     850           846
## # i 709 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>
```

`filter()` all flights in either November or December.

```
flights %>% filter(month == 11 | month == 12)
```

```
## # A tibble: 55,403 × 19
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
## 1  2013    11     1       5           2359           6     352           345
## 2  2013    11     1      35           2250        105     123           2356
## 3  2013    11     1     455           500          -5     641           651
## 4  2013    11     1     539           545          -6     856           827
## 5  2013    11     1     542           545          -3     831           855
## 6  2013    11     1     549           600         -11     912           923
## 7  2013    11     1     550           600         -10     705           659
## 8  2013    11     1     554           600          -6     659           701
## 9  2013    11     1     554           600          -6     826           827
## 10 2013    11     1     554           600          -6     749           751
## # i 55,393 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>
```

`filter()` all flights not delayed by two or more hours.

```
flights %>% filter(!(arr_delay > 120 | dep_delay > 120))
```

```
## # A tibble: 316,050 × 19
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
## 1  2013     1     1     517           515           2     830           819
## 2  2013     1     1     533           529           4     850           830
## 3  2013     1     1     542           540           2     923           850
## 4  2013     1     1     544           545          -1    1004          1022
## 5  2013     1     1     554           600          -6     812           837
## 6  2013     1     1     554           558          -4     740           728
## 7  2013     1     1     555           600          -5     913           854
## 8  2013     1     1     557           600          -3     709           723
## 9  2013     1     1     557           600          -3     838           846
## 10 2013     1     1     558           600          -2     753           745
## # i 316,040 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>
```

`filter()` all flights that departed in summer (July, August, and September).

```
flights %>% filter(month >= 7 & month <= 9)
```

```
## # A tibble: 86,326 × 19
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
## 1  2013     7     1       1           2029          212     236           2359
## 2  2013     7     1       2           2359           3     344           344
## 3  2013     7     1      29           2245          104     151            1
## 4  2013     7     1      43           2130          193     322            14
## 5  2013     7     1      44           2150          174     300           100
## 6  2013     7     1      46           2051          235     304           2358
## 7  2013     7     1      48           2001          287     308           2305
## 8  2013     7     1      58           2155          183     335            43
## 9  2013     7     1     100           2146          194     327            30
## 10 2013     7     1     100           2245          135     337           135
## # i 86,316 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>
```

`filter()` all flights that flew to Houston (IAH or HOU).

```
flights %>% filter((dest == 'IAH' | dest == 'HOU'))
```

```
## # A tibble: 9,313 × 19
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
## 1  2013     1     1     517           515           2     830           819
## 2  2013     1     1     533           529           4     850           830
## 3  2013     1     1     623           627          -4     933           932
## 4  2013     1     1     728           732          -4    1041          1038
## 5  2013     1     1     739           739           0    1104          1038
## 6  2013     1     1     908           908           0    1228          1219
## 7  2013     1     1    1028          1026           2    1350          1339
## 8  2013     1     1    1044          1045          -1    1352          1351
## 9  2013     1     1    1114           900          134    1447          1222
## 10 2013     1     1    1205          1200           5    1503          1505
## # i 9,303 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>
```

## Using `arrange()`

`arrange()` flights by descending order.

```
flights %>% arrange(desc(dep_delay))
```

```
## # A tibble: 336,776 × 19
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
## 1  2013     1     9     641             900         1301    1242         1530
## 2  2013     6    15    1432            1935         1137    1607         2120
## 3  2013     1    10    1121            1635         1126    1239         1810
## 4  2013     9    20    1139            1845         1014    1457         2210
## 5  2013     7    22     845            1600         1005    1044         1815
## 6  2013     4    10    1100            1900          960    1342         2211
## 7  2013     3    17    2321             810          911     135         1020
## 8  2013     6    27     959            1900          899    1236         2226
## 9  2013     7    22    2257             759          898     121         1026
## 10 2013    12     5     756            1700          896    1058         2020
## # i 336,766 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>
```

## Using select()

select() year, month, and day.

```
flights %>% select(year, month, day)
```

```
## # A tibble: 336,776 × 3
##   year month   day
##   <int> <int> <int>
## 1  2013     1     1
## 2  2013     1     1
## 3  2013     1     1
## 4  2013     1     1
## 5  2013     1     1
## 6  2013     1     1
## 7  2013     1     1
## 8  2013     1     1
## 9  2013     1     1
## 10 2013     1     1
## # i 336,766 more rows
```

select() select all columns between year and day (inclusive).

```
flights %>% select(year:day)
```



```
## # A tibble: 336,776 × 3
##   year month   day
##   <int> <int> <int>
## 1  2013     1     1
## 2  2013     1     1
## 3  2013     1     1
## 4  2013     1     1
## 5  2013     1     1
## 6  2013     1     1
## 7  2013     1     1
## 8  2013     1     1
## 9  2013     1     1
## 10 2013     1     1
## # i 336,766 more rows
```

`select()` select all columns except those from year to day (inclusive).

```
flights %>% select(year:day)
```

```
## # A tibble: 336,776 × 3
##   year month   day
##   <int> <int> <int>
## 1  2013     1     1
## 2  2013     1     1
## 3  2013     1     1
## 4  2013     1     1
## 5  2013     1     1
## 6  2013     1     1
## 7  2013     1     1
## 8  2013     1     1
## 9  2013     1     1
## 10 2013     1     1
## # i 336,766 more rows
```

## Using summarise()

`summarise()` by removing NAs prior to computation.

```
flights %>% summarise(delay = mean(dep_delay, na.rm = TRUE))
```

```
## # A tibble: 1 × 1
##   delay
##   <dbl>
## 1  12.6
```

## Using mutate()

Use `mutate()` to create new variables.

```
flights %>% mutate(gain = dep_delay - arr_delay, hours = air_time / 60,
  gain_per_hour = gain / hours)
```

```
## # A tibble: 336,776 × 22
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>       <dbl>   <int>         <int>
## 1  2013     1     1     517           515         2     830           819
## 2  2013     1     1     533           529         4     850           830
## 3  2013     1     1     542           540         2     923           850
## 4  2013     1     1     544           545        -1    1004          1022
## 5  2013     1     1     554           600        -6     812           837
## 6  2013     1     1     554           558        -4     740           728
## 7  2013     1     1     555           600        -5     913           854
## 8  2013     1     1     557           600        -3     709           723
## 9  2013     1     1     557           600        -3     838           846
## 10 2013     1     1     558           600        -2     753           745
## # i 336,766 more rows
## # i 14 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>, gain <dbl>, hours <dbl>,
## #   gain_per_hour <dbl>
```

```
flights %>%
  mutate(gain = dep_delay - arr_delay,
    hours = air_time / 60,
    gain_per_hour = gain / hours,
    fast_plane = ifelse(gain_per_hour > 1, 1, 0))
```

```
## # A tibble: 336,776 × 23
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>       <dbl>   <int>         <int>
## 1  2013     1     1     517           515         2     830           819
## 2  2013     1     1     533           529         4     850           830
## 3  2013     1     1     542           540         2     923           850
## 4  2013     1     1     544           545        -1    1004          1022
## 5  2013     1     1     554           600        -6     812           837
## 6  2013     1     1     554           558        -4     740           728
## 7  2013     1     1     555           600        -5     913           854
## 8  2013     1     1     557           600        -3     709           723
## 9  2013     1     1     557           600        -3     838           846
## 10 2013     1     1     558           600        -2     753           745
## # i 336,766 more rows
## # i 15 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>, gain <dbl>, hours <dbl>,
## #   gain_per_hour <dbl>, fast_plane <dbl>
```

# Descriptive Statistics

This is an introduction to `dplyr` for Faster Data Manipulation in R. There are five basic verbs: `filter()`, `select()`, `arrange()`, `summarise()` (plus `group_by()`), and `mutate()` - Pick observations by their values (`filter()`); - Reorder the rows (`arrange()`); - Pick variables by their names (`select()`); - Create new variables with functions of existing variables (`mutate()`); - Collapse many values down to a single summary (`summarise()`)

These can all be used in conjunction with `group_by()` which changes the scope of each function from operating on the entire dataset to operating on it group-by-group.

```
describe(flights$dep_time)
```

```
##      vars      n    mean      sd median trimmed   mad min  max range  skew
## X1      1 328521 1349.11 488.28   1401 1346.82 634.55    1 2400  2399 -0.02
##      kurtosis    se
## X1      -1.09 0.85
```

```
mean(flights$dep_time, na.rm = TRUE)
```

```
## [1] 1349.11
```

```
median(flights$dep_time, na.rm = TRUE)
```

```
## [1] 1401
```

```
sd(flights$dep_time, na.rm = TRUE)
```

```
## [1] 488.2818
```